Education to Eternity

JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
An Autonomous Institute, with NAAC "A" Grade
Basic Science & Humanities Department
2023-24 (EVEN Semester)
CLASS TEST- I EXAMINATION

Course: B. Tech.
Subject: Advanced Chemistry
Subject Code: CS2T002, AI2T002, DS2T002
Year/Semester: IInd Semester (First Year)
Max Marks: 20 **Date :- 26/02/2024** **Duration:- 1 Hr**

Instructions to the Students:
1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

CO1 Describe various properties of water, Describe types of fuel, refining of Petroleum, orbital ,electronic configuration, nonmaterial and polymers, energy levels, spectroscopic technique.
CO2 Interpret the various classification of fuel, various sources of water , refining of petroleum, classification of CNT, various properties of nonmaterial and polymers, ionization energies.
CO3 Apply the Knowledge of characteristics of good fuel, Synthesis of nonmaterial, liquid crystal polymers, zeolite process, Ion exchange process, Hot Lime –Soda process, acid base concept, spectroscopic techniques.
CO4 Analyze the question on Proximate and Ultimate analysis of coal, potential use of nonmaterial, phases of thermo tropic polymers, analyze question on water characteristics.
CO5 Estimate a Modal on commercial grading of coal, synthesis of nonmaterial, advanced polymers,spectroscopic technique, doping.
CO6 Organize coal, water as per quality, energy level diagram of diatomic molecules, nonmaterial and polymers.

SECTION A
1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	QUESTIONS	Level/CO Mapped	Marks
Q.1	What are the two primary Cations responsible for causing hardness inwater?	1/CO1	1
Q.2	Name the type of hardness that can be removed by boiling water.	1/CO1	1
Q.3	Which type of hardness can be reduced through the addition of certain chemicals like lime and soda?	1/CO3	1
Q.4	Explain permanent hardness and provide an example of a salt contributingto it.	1/CO1	1

SECTION B
1. **All Questions are compulsory.**
2. **Answer in brief.**

Q.5	Why is it important to soften water before using it in industrial processes	2/CO3	2
Q.6	How does the lime-soda process help to do the softening of water?	2/CO3	2
Q.7	Describe the process of ion exchange in the ion-exchange process with its regeneration reaction.	2/CO3	2
Q.8	Can you describe the process of regeneration of zeolite in water treatment?	2/CO3	2

SECTION C
Solve any two questions.

Q.9	Can you evaluate the benefits and drawbacks of using the zeolite process compared to the Lime soda process in terms of cost, efficiency, and environmental impact?	4/CO3	4
Q.10	How would you compare the effectiveness of the lime-soda process and the ion exchange process in treating hard water for industrial use?	3/CO3	4
Q.11	How does the hot lime soda process make water better by dealing with hardness, and what makes it different from the cold methods?	4/CO3	4

$Na_2Z + MgSO_4$ (CaCl2) $\rightarrow$ $MgZ + Na_2SO_4$ (2NaCl)

$MgZ + 2NaCl \rightarrow Na_2Z + MgCl_2$

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